PSP-144 · NQVP EXPANSION SPEC

Enhanced Psychics Framework v3.0

A falsifiable instrumentation and protocol specification for consumer-grade EEG: D^3-axis state estimation, hyper-correlation amplification ladders, and blind-scored Channel / Remote-View trial methodology — engineered as an observation layer for the Ghost Mesh ecosystem.

Ghost Mesh Research Division

Sovereign Hub · Neuro-Quantum Validation Program

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1. Executive Summary and Framework Charter

The Enhanced Psychics Framework v3.0 (EPF-3.0) is an instrumentation and protocol specification that extends the Unified Theory of Degens (UTD v0.3), the Correlation Continuum (CC), and Unified Holographic Gnosis (UHG) into an *operational observation layer*. Where previous generations mapped cognitive states as coordinates in the triadic D^3 space of Precision (*P*), Boundary (*B*), and Temporal (*T*) axes, v3.0 binds those coordinates to measurable electrophysiology: a DIY-constructed EEG acquisition stack, a documented signal-processing pipeline, a five-level amplification ladder, and a blind-scored trial methodology for the experiences the framework conventionally labels “Channel” and “Remote View”. The document is written in falsifiable-spec mode: every operative claim is paired with an estimator, a pre-registered test, and an explicit kill criterion.

> ***Charter caveat.*** *All framework content is theoretical hypothesis generation, not established clinical fact. “Psychic” phenomena are treated as computationally real but metaphysically misinterpreted: above-chance information transfer, where it is measured, is attributed to subtle-cue inference, statistical anticipation, and inter-person coupling — not to paranormal reception. Nothing in this specification asserts that consumer EEG equipment demonstrates, amplifies, or validates paranormal ability. This document is not medical advice, not a clinical instrument, and not a treatment protocol. Research use only.*

The scope of v3.0 covers four pillars selected by the framework's operators: (i) the science and construction of the EEG headset itself, specified as a reproducible DIY build path; (ii) the signal-processing chain that converts raw electrode voltages into D^3 state estimates; (iii) modulation protocols that train targeted states through neurofeedback and structured sessions; and (iv) a Channel / Remote-View protocol layer whose claims are bound to blind judging and pre-agreed statistics. Supporting chapters add dashboard integration hooks for the PSP-144 Sovereign Hub, a safety and load-limit section, and a validation program with pre-registered studies EP-1 through EP-4.

1.1 Version Charter

Version 3.0 is the first release that treats the observing instrument as part of the theory. The charter below situates the release relative to prior framework generations. Each row names the artifact, its contribution, and its status in the current specification. The framework's lineage matters because v3.0 reuses, rather than redefines, the axis algebra, the comorbidity distance metric, and the Coherence Polytope vocabulary established by UTD v0.3 and the PSP-144 NQVP dashboard generation.

Release	Contribution	Status in v3.0
UTD v0.3	Triadic *P*/*B*/*T* axis algebra; disorder atlas; comorbidity distance; dial model	Reused unchanged as the state-space definition
CC / UHG	Relational ontology; coherence conservation; invertibility bounds (noise $\sigma \leq 5.3\%$)	Interpretive layer only; no operative claim in v3.0 depends on it
PSP-144 NQVP v2.0	Ghost Mesh dashboard; D^3 axis monitors; 24 intelligence modules; multi-LLM API constellation	Extended: v3.0 adds real EEG feeds behind the demo metrics
EPF v3.0 (this spec)	EEG observation layer; DIY hardware path; amplification ladder; blind-scored Channel/RV layer; kill criteria	Operative specification; falsifiable mode engaged

Table 1: Release lineage and the status of each inherited layer.

Terminology conventions used throughout: italic *P*, *B*, *T* denote the Precision, Boundary, and Temporal axes; a circumscribed hat (̂) marks an estimated quantity, e.g. P̂ is the EEG-derived estimator of the Precision axis; z denotes a z-score against the operator's own session-0 baseline. “Operator” is the person wearing the headset, “analyst” the person running the pipeline, and “witness” an independent scorer who never sees trial metadata. All numerical thresholds in this document are defaults, not laws: they may be tightened by pre-registration but never loosened after data collection begins.

2. Foundational Architecture: The Observed Mind

The UTD master equation models a cognitive state as a deterministic transform of three axes plus plasticity noise. v3.0 adds what every empirical framework eventually needs: an observation equation. The operator's internal state is never measured directly; the EEG headset records scalp potentials that are a lossy, noise-corrupted projection of that state. Formally, the framework now posits two coupled equations — a state equation inherited from UTD, and an observation equation that defines what the instrument can and cannot see. This separation is the architectural heart of the release, because every estimator, protocol, and kill criterion downstream is a statement about the observation layer, not about the metaphysics of the state space.

$$\Psi(t) = F(\, P(t),\, B(t),\, T(t)\,) + \xi(t) \tag{2.1}$$

$$d(t) = h(\, \Psi(t)\,) + \varepsilon(t) \tag{2.2}$$

$$(\, \hat{P},\, \hat{B},\, \hat{T}\,) = \arg\min_{P,B,T} \| \, d(t) - h(\, F(P,\, B,\, T)\,)\, \|^2 + \lambda R(P,\, B,\, T) \tag{2.3}$$

Equation 2.1 restates the UTD master equation, where $\xi(t)$ is plasticity noise. Equation 2.2 introduces the observation channel $h(\cdot)$ mapping neural state to scalp measurements $d(t)$ — for this specification, an 8-to-16-channel consumer EEG stream sampled at 125–250 Hz — with measurement noise $\varepsilon(t)$ that includes muscle artifact, electrode drift, and environmental interference. Equation 2.3 defines estimation as a regularized inverse problem: the operator's coordinates are the argument that best explains the recording under a smoothness penalty R and regularization strength λ. Because h is many-to-one, the inverse is ill-posed; the specification therefore mandates multi-session baselines, multi-site montages, and explicit confidence reporting rather than single-shot state readings.

The healthy reference state remains $x_0 = (0, 0, 0)$, and state severity remains the Euclidean distance $\|x - x_0\|$. What changes in v3.0 is that the amplification protocols of Chapter 6 and the trial layer of Chapter 7 target *estimator-space* regions, not clinical claims. The framework defines two operational targets. First, **hyper correlation**: measured information transfer above chance that is attributable to specifiable signals — micro-expression pickup, prosodic cues, statistical regularities of shared environments, and inter-person physiological coupling during coordinated action. Second, **the reception posture**: a reproducible operator state in which Precision is high but stabilized, Boundary is moderately softened, and Temporal orientation is balanced; the posture under which subtle-cue performance is predicted to peak. Neither target asserts an unphysical channel; both are bound to measurable outcomes with pre-agreed statistics.

Channel and Remote View are consequently defined as *states and tasks*, not faculties. A Channel session is an operator-held reception posture (estimator band specified in Chapter 7) during a free-response task scored blind. A Remote-View trial is a structured attempt to describe a hidden target under full sensory isolation, judged by independent witnesses against a fixed target pool. The framework's honest base-rate expectation for strictly blinded trials is chance performance; Chapter 7 is written so that a null result is a clean, publishable outcome and any above-chance effect must survive replication before the framework updates its claims.

3. The D^3-EEG Correlate Map

This chapter is the specification's core module: it binds each D^3 axis to a defensible electrophysiological estimator. The mappings are population-level correlations drawn from the cognitive-electrophysiology literature — aberrant salience and gamma-band gain for Precision, default-mode-network coupling for Boundary, and frontal-midline theta for Temporal horizon — and they are calibrated per operator, never interpreted as diagnostic markers. Each estimator is a z-score against the operator's own eyes-open baseline recorded at Level 1 of the training ladder. The chapter defines the estimators formally, then tabulates signatures, sites, and expected directions of change.

3.1 Estimator Definitions

$$\hat{P} = z[\ \log(\ \Gamma_{30\text{–}50} / \alpha_{8\text{–}12}\ )\ ] + 0.5 \cdot z[MMN] \tag{3.1}$$

$$\hat{B} = (\ C_{int} - C_{ext}\ ) / (\ C_{int} + C_{ext}\ ) \tag{3.2}$$

$$\hat{T} = z[\ \text{slope of } Fm\theta_{Fz} \text{ across a delay-discounting task}\ ] \tag{3.3}$$

$$d(A, B) = \sqrt{(\ (\Delta P)^2 + (\Delta B)^2 + (\Delta T)^2\ )} \tag{3.4}$$

Equation 3.1 estimates Precision as the z-scored log ratio of 30–50 Hz gamma power to 8–12 Hz alpha power, augmented by half-weighted mismatch-negativity (MMN) amplitude from an auditory oddball block; the ratio indexes cortical gain on prediction error, the computational quantity the UTD framework calls precision. Equation 3.2 estimates Boundary as the normalized difference between intra-network alpha coherence — mean pairwise coherence among Pz, P3, and P4, the posterior default-mode proxy — and external coupling, taken as Fz–Cz coherence during a working-memory probe; low $\hat{B}$ corresponds to relative dominance of internally oriented coupling. Equation 3.3 estimates the Temporal axis from the slope of frontal-midline theta at Fz as subjective delay lengthens in a discounting task, consistent with Fmθ tracking horizon evaluation. Equation 3.4 reuses the UTD comorbidity distance between estimated states, preserving compatibility with the v0.3 disorder atlas and the dashboard's comorbidity predictions.

3.2 Biomarker Table and Spectral Map

Axis	Primary EEG signature	Sites	Direction at high values
P Precision	Gamma/alpha power ratio; MMN amplitude; gamma phase-lock to prediction error	Fz, Cz, C3/C4	Noise weighted as signal; hypervigilant gain on error responses
B Boundary	Posterior DMN internal alpha coherence vs. DMN–task coupling; TPJ alpha reactivity	Pz, P3, P4 vs. Fz	Rigid self–other demarcation when high; porous coupling when low
T Temporal	Frontal-midline theta (4–8 Hz) slope under delay; CNV amplitude	Fz, FCz	Future-locked horizon when positive; past-lock and rumination signatures when negative
Interaction	Alpha–gamma cross-frequency coupling during binding tasks	Pz–Fz	Indexes Coherence Polytope utilization r/d_s

Table 2: D^3 axes, their EEG estimators, montage sites, and expected directions of change.

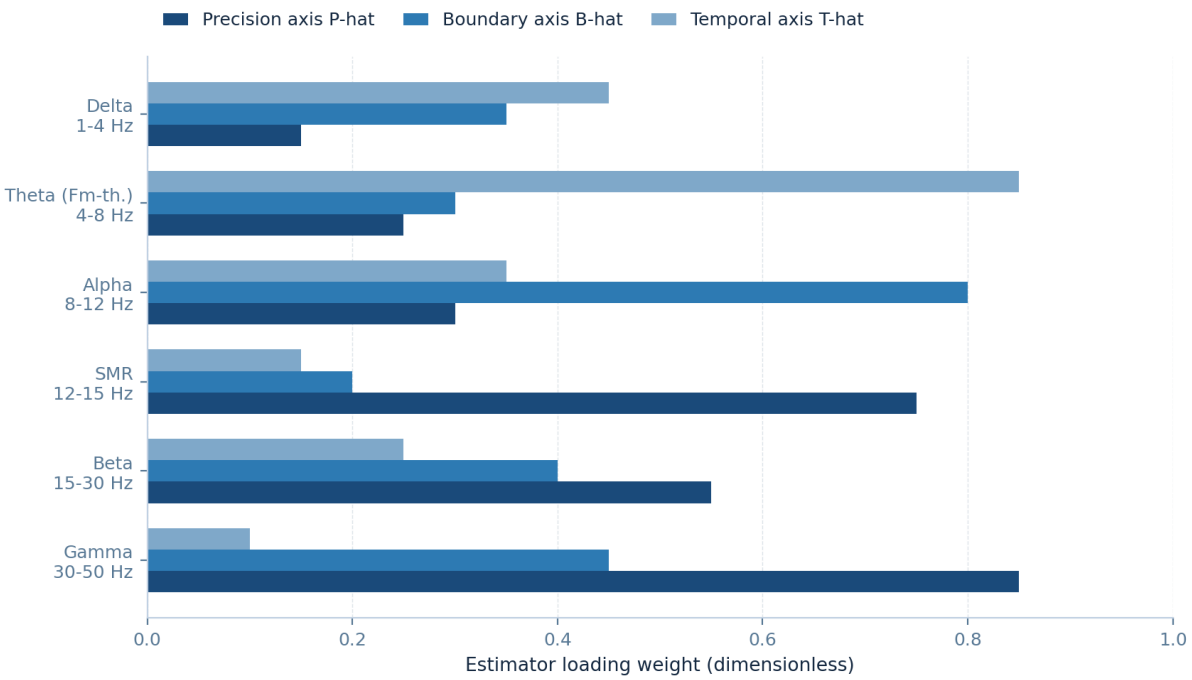


Figure 1: Spectral loading map of the D^3 estimators — model-defined weights linking each band to each axis (Section 3.1). Bars are specification constants subject to revision by EP-1.

Three caveats govern every use of the map. First, gamma estimates above 30 Hz are contaminated by EMG from jaw and neck musculature; the pipeline therefore enforces an EMG screen (Chapter 5) before any gamma-derived precision value is logged. Second, single-operator coherence measures have wide confidence intervals at consumer channel counts; Ƀ is reported with bootstrap confidence bounds and is only interpretable after at least two baseline sessions. Third, the map is a model of correlations, not of mechanisms: EP-1 exists precisely to measure the test-retest reliability of these estimators, and a reliability failure retires the estimator layer without touching the underlying axis algebra.

4. DIY EEG Hardware Build Path

The reference instrument for v3.0 is a passive-recording, battery-operated, 24-bit consumer EEG stack assembled from open-source components. The design goal is not research-laboratory fidelity but a documented, reproducible floor: enough channels, bandwidth, and signal integrity for the estimator layer of Chapter 3 to produce stable z-scores, at a price a single operator can absorb. The build centers on an ADS1299-based amplifier (the OpenBCI Cyton family) mounted in an Ultracortex-style headset, with the 10–20 international montage positions reduced to the eight sites the D^3 estimators actually consume. v3.0 records only: no transcranial stimulation, no photic entrainment hardware, and no device modifications that inject current into the operator.

Component	Specification	Role in the stack
Amplifier board	OpenBCI Cyton — 8 channels, 24-bit ADS1299, 250 Hz sampling, CMRR ≈ −110 dB, built-in bias drive	Primary acquisition; the 8-channel budget maps exactly to the Chapter 3 montage
Extension board	OpenBCI Daisy — +8 channels at 125 Hz aggregate	Optional; enables the 16-channel research configuration for EP-3
Headset frame	Ultracortex Mark IV (printable) or elastic 8-channel cap	Mechanical mounting at 10–20 positions with consistent pressure
Electrodes	Dry polymer (stock) / saline-soaked / gold-cup with paste	Dry for speed of setup; gel upgrade recommended for Fz and Pz (estimator sites)
Reference & bias	A1/A2 ear-clip linked-mastoid reference; SRB2 bias output	Reference defines the measurement; bias drive cancels common-mode interference
Host & link	Battery-powered laptop, USB or BLE dongle, LSL-compatible acquisition software	Isolates the operator from mains; streams to the pipeline

Table 3: Reference DIY build. All parts are commercially available; no soldering to the operator path is required.

4.1 Montage: Eight Channels That Matter

Ch	10–20 site	Estimator role
1	Fp1	Frontal precision probe; alpha-theta training (L3); blink-artifact donor channel for ICA
2	Fp2	Frontal precision probe (right); asymmetric-gain check on P
3	Fz	Frontal-midline theta for T; SMR/theta neurofeedback reference for L2-L3
4	Cz	SMR (12–15 Hz) primary training site at C3/C4–Cz derivations
5	C3	Sensorimotor rhythm left; motor imagery trials in the Channel battery
6	C4	Sensorimotor rhythm right; symmetry control for C3
7	Pz	Posterior DMN hub for B internal-coherence term; alpha posterior dominant rhythm
8	Oz	Posterior alpha anchor; eyes-closed PDR verification; visual target processing marker

Table 4: Framework montage. Linked A1/A2 ear clips form the reference; SRB2 is the bias. P3/P4 derive from Pz neighbors when the Daisy extension is fitted.

Impedance discipline is the difference between a spec and a toy. Dry polymer electrodes should read at or below roughly 50 kΩ at the skin after seating and misting; saline cups below 25 kΩ; pasted gold cups below 10 kΩ. The stack's impedance-check mode is run at the start of every session and the values are logged into the session JSON (Chapter 8) so the analyst can weight the estimate accordingly. Motion artifact is handled procedurally rather than electronically: the operator's jaw and neck are scanned for EMG before gamma values are trusted, long cables are dressed away from the face, and the operator stays on battery power with the laptop lid open at arm's length — never connected to a mains-powered charger, audio amplifier, or any

stimulation device during a recording.

4.2 Cost Ladder

Tier	Configuration	Approx. cost	Capable of
0	Muse-class headband: 4 dry channels (TP9, AF7, AF8, TP10), 4–5 electrode sites	≈ $250	Alpha training and L1 mapping only; insufficient for B coherence
1	4-channel ganglion-class board + 8-position dry headset	≈ $500	L1-L2; partial estimator coverage
2 (recommended)	Cyton 8ch + Mark IV + saline cups + battery pack	≈ $1,100–1,300	Full Chapter 3 montage; L1-L5
3	Tier 2 + Daisy 16ch + gold-cup gel cap upgrade	≈ $1,600+	Research-grade SNR; required for multi-site EP-3 hyperscanning replication

Table 5: Build tiers. Tier 2 is the recommended specification for operators following the full ladder.

8 ch · 24-bit · 250 Hz

Tier 2 recommended spec — the minimum configuration that covers every estimator in Chapter 3

Two safety rules are absolute for the DIY path. First, the acquisition chain must be battery-isolated from mains at all times during a session; laptop chargers, desktop towers, and unisolated USB hubs are prohibited on the operator side. Second, the electrode path is passive: no build may add current injection of any kind. Skin irritation from paste or pressure is managed with session-length caps (Chapter 9), and the headset is removed at any sign of discomfort. The completed build is an observation instrument, not a therapeutic device, and it makes no medical claims.

5. Signal Processing Pipeline

Raw scalp voltage becomes a state estimate through a fixed, version-pinned pipeline. The default chain is: import and event annotation; re-referencing to linked mastoids (with average reference computed for cross-checks); zero-phase band-pass 0.5–50 Hz; notch at 50 or 60 Hz matched to local mains; independent component analysis to remove blink, saccade, and jaw-EMG components; epoching around task events; and spectral estimation by Welch's method with 2-second windows and 50% overlap. Every parameter in this chain is recorded into the session JSON so that any published statistic can be recomputed exactly. The pipeline is deliberately boring: reproducibility is the amplification framework's first defense against fooling itself.

Band	Range	Role in the framework	Primary sites
Delta	1–4 Hz	Load indicator; elevated delta fragments estimation quality	All
Theta	4–8 Hz	Frontal-midline theta for T; theta/SMR balance in L2	Fz, FCz
Alpha	8–12 Hz	Boundary estimator substrate; posterior dominant rhythm; L3 carrier	Pz, P3, P4, Oz
SMR	12–15 Hz	Precision stabilization target in L2; thalamocortical idling	C3, Cz, C4
Beta	15–30 Hz	Active-processing marker; suppression check during SMR training	C3, C4
Gamma	30–50 Hz	Precision gain term (Eq. 3.1); mandatory EMG screen	Fz, Cz

Table 6: Band architecture and the role of each range in the estimator and protocol layers.

5.1 Core Metrics

$$S_{xx}(f) = (1/K) \sum_{k=1..K} \|X_k(f)\|^2 \tag{5.1}$$

$$P_{band} = \int_{f1}^{f2} S_{xx}(f)\, df \tag{5.2}$$

$$C_{xy}(f) = \|S_{xy}(f)\|^2 / (S_{xx}(f) \cdot S_{yy}(f)) \tag{5.3}$$

$$PLV = \| (1/N) \sum_{n=1..N} e^{i(\varphi_x(n) - \varphi_y(n))} \| \tag{5.4}$$

$$HCI = w_1 \cdot PLV_{dyad} + w_2 \cdot r_{behavior} + w_3 \cdot (1 - \sigma_{noise}),\ w = (0.5, 0.3, 0.2) \tag{5.5}$$

Equations 5.1 and 5.2 define band power as the integral of the averaged periodogram over the band edges of Table 6; this is the quantity every estimator z-score consumes. Equation 5.3 is magnitude-squared coherence between channel pairs, the substrate of the Boundary estimator's internal and external coupling terms, evaluated per frequency and averaged over the alpha band. Equation 5.4 is the phase-locking value — the mean resultant length of inter-channel phase differences — which serves both as the dyadic synchrony metric of Level 4 and, between two simultaneously recorded operators, as the Ghost Mesh coupling measure of Chapter 7. Equation 5.5 composes the Hyper-Correlation Index from the dyadic PLV lift, the blind-judged behavioral transfer correlation, and a noise-inquality term; the default weights are specification constants, and HCI is always reported with a bootstrap confidence interval rather than as a bare number.

Statistical hygiene is part of the pipeline, not an afterthought. Coherence and PLV claims are tested against within-operator surrogate distributions generated by phase randomization, with 1,000 permutations per contrast. Comparisons across the eight montage sites or six bands use Holm-corrected thresholds. Gamma-derived precision values are voided whenever the EMG screen's high-frequency floor (60–80 Hz) rises more than two standard deviations above the session baseline. These rules exist so that the amplification ladder's successes and failures are both trustworthy.

6. Hyper-Correlation Amplification Protocols

Amplification, as this specification uses the word, is the training of a state, not the strengthening of a signal. The mechanism is grounded: human performance on implicit information-transfer tasks — reading micro-expressions, anticipating a partner's moves, sensing prosodic stress before words resolve — varies measurably with cortical state, attentional gain, and anxiety load. In D^3 terms, the framework predicts peak transfer when Precision is high but stable (not chaotic), Boundary is flexible enough to model the other person without dissolving, and Temporal orientation is balanced. The five-level ladder below trains exactly that posture, one axis at a time, with neurofeedback providing the mirror. Every level's exit criterion is an estimator-space statement, so progression is auditable rather than felt.

Level	Protocol	EEG target	Sessions	Exit criterion
L1	Baseline mapping: 10-min eyes-open and eyes-closed blocks; auditory oddball for MMN; discounting task	Individual band norms; Ƥ, Ɓ, Ƭ baselines	2	Baseline z-scores stable within 0.5 SD across both sessions
L2	Precision stabilization: SMR (12–15 Hz) uptraining at Cz with theta not suppressed; threshold feedback	SMR z ≥ +1.0 sustained ≥ 60% of session	12	SMR ratio gain ≥ 1.5 z vs. L1, replicated in 2 consecutive sessions
L3	Boundary modulation: alpha–theta crossover at Fz/Pz; guided entry/exit scripts	Controlled Ɓ drop of 0.5–1.0 with Ƥ variance below threshold	8	Reproducible state entry 3× with operator-logged lucidity of exit
L4	Dyadic coherence lock: paired 0.1-Hz breathing, joint-attention block, dual-headset recording	Alpha-PLV lift ≥ 2 SD vs. each operator's solo baseline	8	PLV lift present in 2 independent dyads under blinding
L5	Blind transfer trials: Chapter 7 target battery under full protocol	HCI at or above trained threshold	ongoing	Pre-registered EP-4 aggregate evaluation at N = 100 sessions

Table 7: The five-level amplification ladder. Session caps and contraindications are governed by Chapter 9.

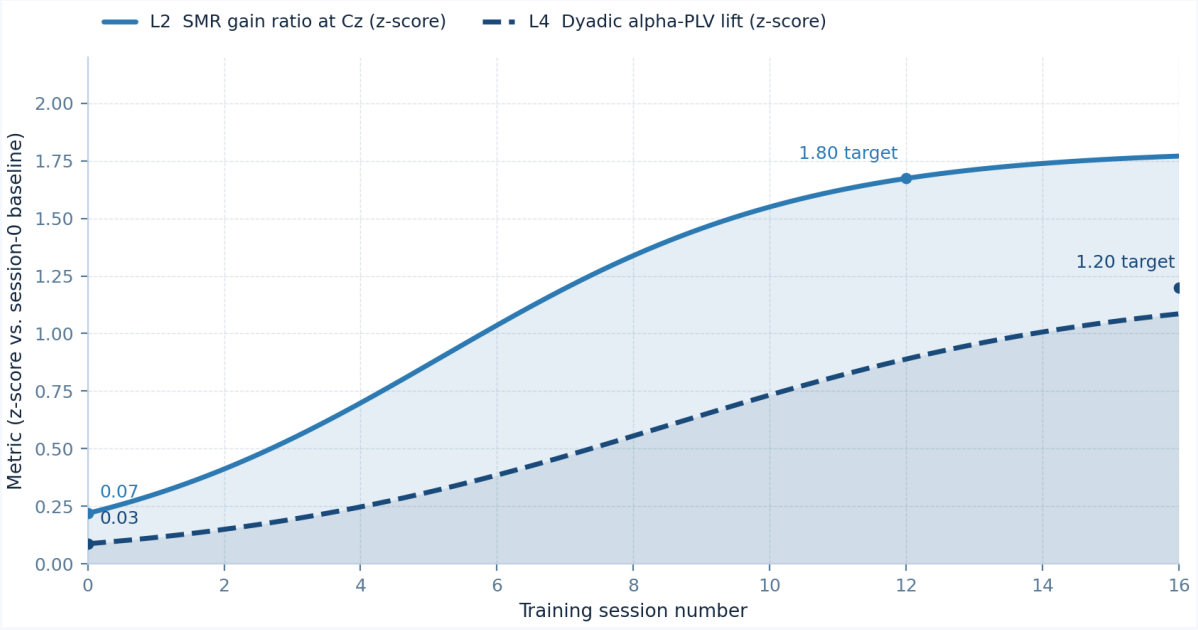


Figure 2: Model target trajectories for L2 and L4 metrics across training sessions — these are pre-registered predictions to be falsified by EP-2 and EP-3, not measured data.

6.1 Session Structure and Load Accounting

A standard session runs thirty minutes: five minutes of setup and impedance verification, twenty minutes of protocol, and five minutes of recovery logging in which the operator records subjective state against a fixed checklist. The recovery block is not optional decoration; it is where destabilization shows up first, and Chapter 9's stop rules key on it. Load is accounted with the framework's allostatic integral: the session accumulates effort proportional to how far the operator is pushed from personal baseline on each axis, and the weekly budget caps the total.

$$L_{session} = \int_{0}^{T} \left(\lVert \hat{P} - P_{0} \rVert + \lVert \hat{B} - B_{0} \rVert + \lVert \hat{T} - T_{0} \rVert \right) dt \le \Lambda \tag{6.1}$$

The cap Λ is individualized at L1, but the default policy is three sessions per week, no more than forty-five minutes of headset wear per day, and a hard twenty-minute ceiling on alpha–theta work per session. Protocol leveling is sequential: an operator may not begin L3 before the L2 exit criterion is met, may not enter L4 without a second consenting operator and a witness, and may not generate L5 trial data until both prior exits are logged in the dashboard. Skipping levels is the most common cause of noisy, uninterpretable runs, because the estimator baselines that make later levels meaningful simply do not exist yet.

7. Channel and Remote-View Protocol Layer

The Channel state is defined operationally, by estimator band and by behavior, so that two analysts can agree on whether it occurred. Entry requires: B in the range −1.5 to −0.5 relative to the operator's own baseline (a controlled softening, not dissolution); P held between +0.5 and +1.5 with within-epoch variance below the operator's L2 threshold; and T within ±0.5 of zero. Duration is capped at twenty minutes, a second person is present as monitor for every entry, and exit is scripted and rehearsed at L3. The state is a trained posture, reached through the ladder; it is never induced pharmacologically, and the framework treats any stimulant-assisted boundary softening as an automatic protocol violation under Chapter 9.

7.1 The Trial Battery

Two task families run during Channel-state and Remote-View sessions. The free-response battery follows the gestalt format: the operator records verbal descriptors and ideometric sketches against a target known only by coordinate code, and a witness pool of at least three independent judges ranks all pool responses against the target blind, without access to session metadata. The forced-choice battery uses a standardized target pool of at least one hundred images spanning distinct semantic clusters; the target is selected after session lock by a cryptographically seeded random process, the operator's transcript is sealed with a hash before target reveal, and judging is four-alternative forced choice by a judge who never met the operator. Both families record dual-EEG when a sender is present, making the Ghost Mesh PLV feed a manipulation check on whether the dyad actually coupled.

$$z_{session} = (H - N \cdot p_0) / \sqrt{(N \cdot p_0 \cdot (1 - p_0))},\ p_0 = 0.25 \tag{7.1}$$

$$z_{agg} = \Sigma_{i=1..k}\, z_i / \sqrt{k} \tag{7.2}$$

Statistic	Definition	Threshold	Interpretation
Per-session hit score	H hits in N forced-choice trials (Eq. 7.1)	report only	Descriptive; sessions are not cherry-picked
Stouffer aggregate	z_{agg} across k sessions (Eq. 7.2)	≥ +1.64 at N = 100	Single pre-set success gate for EP-4
Effect size	Cohen's h on hit proportion vs. p_0	reported always	Precision of the null matters as much as the alternative
Multiplicity	Holm correction across pre-named contrasts	$\alpha = 0.05$ family-wise	Protects against montage- and band-shopping
Stopping rule	Fixed N = 100; no interim peeking; deviation voids the run	mandatory	Optional stopping is the classic self-deception engine

Table 8: Scoring statistics for the blind trial layer. All thresholds are fixed before the first trial.

> ***Expected-null clause.*** *Under strict sensory isolation and double-blind judging, the honest base-rate expectation of this framework is chance performance; that is what the Current Best Evidence prior predicts, and it is what most carefully blinded historical programs reported. A null aggregate at N = 100 is a clean, publishable outcome that retires the Channel/RV claim for this configuration (Chapter 10). Any effect observed without blinding, fixed targets, or sealed scoring is anecdote by definition and carries zero evidential weight inside this framework.*

The dyadic session extends the battery to two headsets. The sender views the target; the receiver, isolated in a separate room or behind opaque shielding with auditory masking, free-responds; both streams are recorded on synchronized clocks via the dashboard's LSL bridge. Inter-brain alpha-PLV during the transfer window is compared against each operator's solo baseline and against a non-coordinated control condition in which the sender views a scrambled target. The framework's prediction is directional and modest — coordinated dyads show synchrony lift, and synchrony lift correlates with behavioral transfer scores — and EP-3 tests exactly that prediction with dyads randomized to coordinated and non-coordinated conditions.

8. PSP-144 Dashboard Integration

v3.0 instruments plug into the existing Sovereign Hub rather than building a parallel console. The acquisition host runs a lightweight local service that bridges the LSL acquisition stream to the hub's PHP layer: sessions post to the hub's proxy with the schema below, the hub's D^3 axis monitors bind to live state estimates, the Ghost Mesh canvas renders the PLV feed, and trial hashes land in the same audit log the API constellation already uses. Neural streams are sensitive personal data: the default posture is local-only storage, node identifiers are anonymized at the bridge, and nothing leaves the operator's network unless the operator explicitly enables sharing.

SESSION JSON SCHEMA - epf3.session/v1 (posted with POST /api/eeg/session)

```
{
  "session_id": "epf3-20261019-0930-a1",
  "schema": "epf3.session/v1",
  "device": {
    "board": "cyton-8ch", "fs_hz": 250,
    "bits": 24, "montage": ["Fp1","Fp2","Fz","Cz","C3","C4","Pz","Oz"],
    "reference": "A1A2-linked", "bias": "SRB2",
    "impedance_kohm": [7.2, 8.1, 6.4, 9.0, 12.3, 11.7, 8.8, 15.2]
  },
  "d3_estimate": {"P_hat": 0.83, "B_hat": -0.91, "T_hat": 0.12,
    "ci95": [0.21, 0.18, 0.25], "state": "channel-window"},
  "bands": {"delta": 1.9, "theta": 4.2, "alpha": 8.9,
    "smr": 3.1, "beta": 2.7, "gamma": 0.9},
  "coherence": {"C_int": 0.61, "C_ext": 0.22,
    "plv_dyad": 0.38, "surrogate_p": 0.012},
  "hci": {"value": 0.57, "ci95": [0.41, 0.70]},
  "trials": [{"trial_id": "t-041", "pool": "std-100",
    "target_hash": "sha256:...", "sealed_transcript": "sha256:...",
    "judge_rank": 2, "z_session": 0.58}],
  "safety_flags": ["load-under-Lambda", "emg-screen-pass"],
  "pipeline_version": "epf3.sig/v1.0"
}
```

Method	Endpoint	Purpose
POST	/api/eeg/session	Ingest a completed session JSON; validates schema, impedance floors, and pipeline version; returns audit hash
GET	/api/eeg/state/current	Live D^3 estimate + band powers for the hub's axis monitors (P, B, T panels)
POST	/api/rv/trial	Register a sealed trial: target hash, transcript hash, judge ranking; emits the stopping-rule ledger entry
GET	/api/mesh/plv	Dyadic PLV feed consumed by the Ghost Mesh canvas and Empathy Weaver module
WS	/ws/mesh	Push channel for live mesh coherence; falls back to 1 Hz polling of /api/mesh/plv

Table 9: Integration endpoints exposed by the bridge service.

PHP WIRING - Sovereign Hub hooks (v3.0)

```
// index.php - add the page
$valid_pages = ['hub','crucible','nexus','community','api-forge',
  'axioms','shortcomings','eeg'];

// proxy.php - local bridge as a provider (no external API key)
case 'eeg-hub':
  $ch = curl_init('http://127.0.0.1:8878' . $path);
  curl_setopt_array($ch, [CURLOPT_RETURNTRANSFER => true,
    CURLOPT_POST => true, CURLOPT_POSTFIELDS => $body,
    CURLOPT_TIMEOUT => 5]);
  return json_decode(curl_exec($ch), true) ?: ['error' => 'bridge down'];
```

On the dashboard side, hub.php binds its three axis monitors to `/api/eeg/state/current` so the demo metrics become live values when a bridge is connected, and falls back to the v2.0 placeholders when it is not. The Crucible page gains an L1–L5 progress panel driven by exit-criterion fields in the session log, and the api-forge page lists the bridge as a local constellation entry. All wiring is additive: no existing hub route changes behavior unless an EPF bridge answers on 127.0.0.1.

9. Safety, Ethics and Load Limits

The framework's own axis model generates its safety rules. Boundary-softening work on an already-porous system, or precision-boosting on an already-hyper-precise one, is exactly the manipulation the UTD atlas associates with destabilization in the profiles it maps. The table below therefore keys contraindications to axes, not to folklore. Every operator completes an intake self-screen against this table before L1, re-screens monthly, and logs adverse events in the recovery block. None of this is medical screening; where any doubt exists, the rule is to obtain professional clearance before proceeding.

Condition / context	Risk axis	Rule
Psychosis-spectrum or manic history	*P* gain; *B* dissolution	Clinical oversight required; L3+ boundary work excluded
Current stimulant use (including high-dose caffeine)	*P*↑, *B*↓ stacking	No L3/L4 within 24 h of stimulant exposure
Seizure history	Cortical excitability	Exclude from all entrainment protocols; neurofeedback only with clearance
Severe dissociative symptoms	*B* intolerance	Grounding-first track; no alpha–theta until cleared
Acute sleep deprivation	Estimator noise; *T* destabilization	Reschedule; sessions void if sleep < 6 h
Minors	Consent and developing *B*	Excluded from the entire ladder

Table 10: Axis-keyed contraindications and rules.

3 / week · 45 min · 20 min

Default load caps: sessions per week, daily headset wear, and the hard alpha-theta ceiling per session

Stop rules are mechanical. If the recovery block records derealization lasting more than thirty minutes, intrusive imagery that disrupts sleep, mood collapse, or a Ƀ drop that fails to recover within the session's final five minutes, the ladder halts at the current level and the incident is logged as an adverse event. Two adverse events at the same level retire that level for that operator permanently. A standing crisis referral clause applies: any operator experiencing sustained distress, dissociation, or suicidal ideation is directed to licensed professional care, and the framework is explicit that it offers no crisis intervention of any kind.

Ethics clauses mirror the falsifiable charter. No covert testing: every dyad member consents to the session's structure, and witnesses consent to scoring. No commerce on unproven claims: the framework may not be packaged as psychic services, diagnosis, or therapy while in falsifiable mode. Data dignity: neural recordings stay local by default, sharing is opt-in per session, and shared bundles strip identity beyond an anonymized node ID. Publication symmetry: null results from EP-4 are as publishable within the community as positive ones, and selective reporting voids a program's standing in the mesh.

10. Validation Protocol and Falsifiability Criteria

Four pre-registered studies carry the entire evidential weight of v3.0. Each is designed so that its failure mode is unambiguous: a missed prediction retires a named layer, not the framework's feelings. Registration happens before the first data session — hypothesis, N, thresholds, and analysis code pinned in a public registry entry — and the dashboard's audit hash chain timestamps every session against that registration. Nothing in the table below is exploratory post hoc; anything discovered outside these gates is labeled exploratory and requires fresh registration to become a claim.

ID	Hypothesis	Design	N	Gate
EP-1	Estimator reliability: test-retest ICC of (P, B, T) over 2 weeks ≥ 0.6	Two baseline repeats per operator, both tiers	30	ICC ≥ 0.6 per axis
EP-2	L2 efficacy: SMR neurofeedback beat sham feedback on SMR gain	Randomized, sham-controlled, analyst-blinded	40	d ≥ 0.4
EP-3	Dyadic synchrony: alpha-PLV lift in coordinated vs. non-coordinated dyads ≥ 2 SD	Within-dyad crossover, order counterbalanced	20 dyads	Lift ≥ 2 SD, $p <$ 0.05 corrected
EP-4	Blind transfer: aggregate hit performance above chance ($p_0 = 0.25$)	Fixed N protocol, sealed scoring, public ledger	100 sessions	$z_{agg} \geq +1.64$

Table 11: Pre-registered validation gates for v3.0.

10.1 Kill Criteria

Each gate maps to a named retirement. If EP-1 falls below ICC 0.4 on any axis, that axis's estimator is retired and the axis returns to being purely theoretical vocabulary — the protocol ladder cannot run without it. If EP-2 shows $d < 0.1$, the L2 protocol is retired as ineffective and L3+ entry is frozen until a replacement stabilization protocol registers. If EP-3 shows no synchrony lift, all dyadic claims reduce to behavioral coordination and the Ghost Mesh PLV feed is re-labeled as visualization only. If EP-4 returns z_{agg} below +1.64 at N = 100, the Channel/Remote-View layer is marked *no evidence — exploratory only*, its dashboard modules are disabled by default, and the framework's public materials must state that plainly. These are commitments made before data, in the open, and they are the difference between a falsifiable spec and a belief system with a dashboard.

> *From chaos, through computation, toward evidence. A framework that cannot name the observation that would kill it has already failed; v3.0 names four.*

Version status: EPF-3.0 enters falsifiable mode as of this document. The recommended path for a new operator group is Tier 2 build, EP-1 registration, and L1–L2 execution while the estimator layer earns or loses its keep. Amendments follow semver with two hard rules: thresholds may only tighten within a registered run, and any post-hoc change to a gate requires a new registration and voids the affected program's prior data. The mesh remains open to replication — independent groups running the same pinned pipeline are the only reviewers whose verdicts count.